

Optimal Test Profile in the Context of Software Cybernetics¹

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Abstract

Software cybernetics explores the interplay between software theory/engineering and control theory/engineering. Following the idea of software cybernetics, the controlled Markov chains (CMC) approach to software testing treats software testing as a control problem. The software under test serves as a controlled object, and the (optimal) testing strategy determined by the theory of controlled Markov chains serves as a controller. The software under test and the corresponding (optimal) testing strategy constitute a closed-loop feedback system, and the software state transitions behave as a Markov chain. This paper analyzes the behavior of the corresponding optimal test profile determined by the CMC approach to software testing. It is shown that in some cases the optimal test profile is Markovian, whereas in some other cases the optimal test profile demonstrates a different scenario. The analyses presented in the paper deepen our understanding of the CMC approach to software testing and are related to software operational profile modeling.

1. Introduction

The idea of software cybernetics was first proposed by the author in 1994 with an attempt to apply cybernetic or control-theoretical approaches to solve software engineering problem [1]. If we consider the potential of applying approaches in the area of software theory or theoretical computer science to solve control engineering problem, then we can define software cybernetics as the interplay between software theory/engineering and control theory/engineering. The feasibility and benefits of the idea of software cybernetics are justified when we

treat software testing as a control problem and propose a controlled Markov chains (CMC) approach to software testing [2]. With the CMC approach, the software under test serves as the controlled object modeled as a controlled Markov chain, whereas the software testing strategy serves as the corresponding controlled object. The software under test and the corresponding testing strategy make up a closed-loop feedback system. Unlike conventional approaches to software testing (e.g., branchtesting, functional testing) [3], which are applied to the software under test without an explicit optimization goal, the CMC approach designs an optimal testing strategy to achieve an explicit optimization goal given a priori. Test cases are selected or generated by the optimal testing strategy and thus constitute an optimal test profile. An interesting but unsolved question is whether the optimal test profile makes up a Markov chain, although the corresponding behavior of the software under the optimal testing strategy fits a Markov chain [2]. This question defines the topic of this paper.

Section 2 treats software testing as a control problem. Section 3 reviews the CMC approach to software testing. Section 4 discusses when optimal test profile makes up a Markov chain. Concluding remarks are contained in Section 5.

2. Software Testing as a Control Problem

In the process of software testing, test cases are selected in accordance with a given testing strategy and applied to the software under test. Some test cases reveal failures. The corresponding failure-causing defects are then removed and thus the underlying software states undergo changes. Some test cases don't reveal failures and no state transitions happen to the software under test.

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So uncertainty is associated with the behavior of the software under test. If we treat the corresponding testing strategy as a control policy or controller, then we can treat the software under test as an uncertain controlled object. Further, if an optimization (testing) goal is given explicitly and a priori, then the test data selection becomes as an optimal control problem. In an optimal control problem, a dynamical system (controlled object) is given whose behavior not only follows its own 'laws of motion' (e.g., Markov state transition laws), but may be influenced or regulated by a suitable choice of some of the system's variables, which are called control or action variables. The controls that can be applied at any given time are chosen according to control policies or laws that

are derived from histories of the system and the given performance criterion or performance index. The performance criterion measures or evaluates in some sense that system's response to the control policies being used. The control policies (or controller) and the controlled system constitute a closed-loop feedback system as depicted in Figure 2.1. Then the optimal control problem is to determine a control policy that optimizes (i.e., either minimizes or maximizes) the performance criterion. Therefore in order to solve an optimal control problem, we need to identify three components: a model for the controlled object, a set of admissible control policies, and a performance index (or objective function).

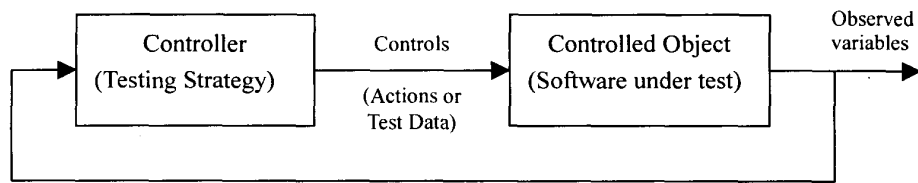


Figure 2.1. Software Testing as a Control Problem.

For the software testing problem of concern, let²

$Y_t = j$; if the software under test contains j defects at time t , $j = 0, 1, 2, \dots, N$; $t = 0, 1, 2, \dots$

$Z_t = \begin{cases} 1 & \text{if the action taken at time } t \text{ detects a defect} \\ 0 & \text{if the action taken at time } t \text{ doesn't detect a defect} \end{cases}$ We

have the following assumptions.

- (1). The software contains N defects at the beginning ($t = 0$).
- (2). An action taken at one time detects at most one defect.
- (3). If a defect is detected, then it is removed immediately and no new defects are introduced; that is, $Y_t = j$ and $Z_t = 1$ mean $Y_{t+1} = j - 1$.
- (4). If no defect is detected, then the number of remaining software defects remains unchanged; that is, $Y_t = j$ and $Z_t = 0$ mean $Y_{t+1} = j$.
- (5). $Y_t = 0$ is an absorbing state; it is the target state.
- (6). At every time there are always m admissible actions; the action set is $A = \{1, 2, \dots, m\}$.
- (7). Z_t depends only on the software state Y_t and the action A_t taken at time t ;

- (8). Action A_t taken at time t incurs a cost of $W_{Y_t}(A_t)$, no matter whether or not it detects a defect;

$$\Pr\{Z_t = 1 | Y_t = j, A_t = i\} = j\theta_i$$

$$\Pr\{Z_t = 0 | Y_t = j, A_t = i\} = 1 - j\theta_i \quad ; j = 1, 2, \dots, N$$

$$\Pr\{Z_t = 0 | Y_t = 0, A_t = i\} = 1 \quad ; t = 0, 1, 2, \dots$$

$$W_{Y_t}(A_t = i) = \begin{cases} w_j(i) & \text{if } Y_t = j \neq 0 \\ 0 & \text{if } Y_t = 0 \end{cases}$$

- (9). The cost of removing a detected defect is ignored³. Let τ be the first-passage time⁴ to state 0, and

$$J_\omega(N) = E_\omega \sum_{t=0}^{\tau} W_{Y_t}(A_t) \quad (2.1)$$

where ω denotes a control policy (testing strategy). Our problem is to find the control policy (testing strategy) that minimizes $J_\omega(N)$. Such a policy removes all the N defects at the least expected cost.

Following the above assumptions, we can use Figure

² An action can be a specific testing technique, a software run, a test case, an equivalence class of test cases, and so forth. It should be interpreted widely.

³ This assumption is not necessary for the subsequent mathematical treatment. However it makes the set of assumptions fit realistic software testing processes.

⁴ The first-passage time to state 0 is the minimal number of actions that are required to move the software under test to state 0. The control (testing) problem is formulated in the context of discrete-time domain. We note that in software reliability modeling, one needs to distinguish between continuous-time domain and discrete-time domain [4].

2.2 to depict the software state transition behavior under test. If action A_t is applied to the software in state j ($j \neq 0$) at time t , then the software remains in state j with probability $1 - j\theta_{A_t}$ and moves to state $j-1$ with probability $j\theta_{A_t}$. Upon entering into the target state 0, the software stays there forever.

3. The Controlled Markov Chains (CMC) Approach to Software Testing

3.1. Controlled Markov Chains

In the setting of controlled Markov chains, the controlled object is given as a discrete-time Markov control model that is five-tuple

$$\langle S, A, \{A(y) | y \in S\}, Q, W \rangle \quad (3.1)$$

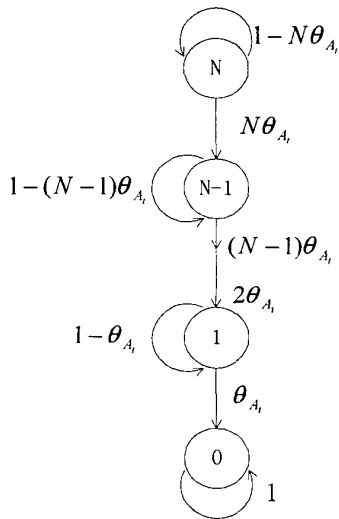


Figure 2.2. Software State Transition under Test.

Where S and A are given sets, called the state space and the control (action) set, respectively. $\{A(y) | y \in S\}$ is a family of nonempty subsets $A(y)$ of A , with $A(y)$ being the set of feasible controls or actions in the state $y \in S$. Q is a transition law such that

$$Q(B | y, a) = \Pr\{Y_{t+1} \in B | Y_t = y, A_t = a\}, B \subset S$$

where Y_t denotes the system state at time t , and A_t the action (control) applied at time t . Finally, W is a cost-per-stage (or one-stage cost) function. When S is discrete, model (3.1) is the so-called controlled Markov

chain.

Control model (3.1) represents a controlled stochastic system that is observed at times $t = 0, 1, 2, \dots$. If the system is in the state $Y_t = y \in S$ at time t and the control $A_t = a \in A(y)$ is applied, then two things happen: (1). A cost $W_t(a)$ is incurred, and (2). The system moves to the next state Y_{t+1} according to the transition law Q . Once the transition into the new state has occurred, a new control is chosen and the process is repeated. Obviously, a controlled Markov chain or discrete-time Markov control model is not a Markov chain in general. It reduces to an ordinary Markov chain when A consists of only one action and $A(y) \equiv A$.

$$\text{Let } H_t = \{Y_0, A_0, Y_1, A_1, \dots, Y_{t-1}, A_{t-1}, Y_t\}$$

That is, H_t denotes the history of the system up to time t .

$$\text{Let } D_t(A_t; H_t) = \Pr\{A_t | H_t\} \text{ i.e., } D_t(a; h_t) \text{ denotes}$$

the conditional probability of $A_t = a$ at time t under the condition⁵ $H_t = h_t$. Obviously,

$$\sum_{a \in A(y_t)} D_t(a; h_t) = 1$$

Then a control policy is defined as

$$\omega = \{D_t(A_t; H_t); t = 0, 1, 2, \dots\}$$

If D_t is always confined to 1 or 0, that is, at any time a particular action is certainly taken, then ω is deterministic⁶. ω is randomized if it is not deterministic. If $D_t(A_t; H_t)$ depends only on t, Y_t and A_t , then the corresponding control policy is Markovian. If $D_t(A_t; H_t)$ depends only on Y_t and A_t , then the corresponding control policy is stationary Markovian (stationary for brevity). In this way we have several classes of control policies: randomized policy, (randomized) Markov policy, (randomized) stationary policy, deterministic policy, deterministic Markov policy, and deterministic stationary policy. Under stationary control policies, we can easily see that the resulting $\{Y_t; t = 0, 1, 2, \dots\}$ forms a (time-homogenous) Markov chain. But this is not true under a general control policy.

Let $W_t(A_t)$ denote the cost incurred at time t by action A_t applied to the system being in state Y_t . Obviously $W_t(A_t)$ is a random variable in general. In the optimal control of a controlled Markov chain we need to optimize certain performance criterion involving

⁵ In the context of software testing we can treat D_t as a moving or time-varying test profile at time t .

⁶ Inconsistent uses of terminology can be observed in the literature [5, 6]. In reference [5] 'deterministic' actually means 'stationary deterministic' here.

$W_{Y_t}(A_t)$. In the literature [5, 6], several classes of performance criteria are proposed, including average cost criteria, first-passage cost criteria, and discounted cost criteria, among others. Here we are most interested in the first-passage cost criteria or the first-passage problem. Suppose the system begins with state $Y_0 = y_0$ and has a target state $y^* \neq y_0$ where the system evolution process should be stopped. Let τ denote the smallest positive integer such that $Y_\tau = y^*$. The optimal first-passage problem is concerned with minimizing the objective function

$$J_\omega(y_0) = E_\omega \sum_{t=0}^{\tau} W_{Y_t}(A_t) \\ = \sum_{t=0}^{\tau} \sum_y \sum_a \Pr^{(\omega)}\{Y_t = y, A_t = a\} W_y(a)$$

with respect to control policy ω , where E_ω denotes the mathematical expectation with respect to control policy ω , $\Pr^{(\omega)}$ denotes the corresponding policy under control policy ω . Here we note that τ is a random variable and therefore $J_\omega(y_0)$ is the expected value of a random sum of random variables.

The following lemma is basic in the theory of controlled Markov chains [5]. It asserts that the solution to the optimal first-passage problem is a deterministic stationary control policy.

Lemma 3.1 (1). If $\{W_{Y_t}(A_t), t = 0, 1, 2, \dots\}$ are nonnegative, then there exists a deterministic stationary control policy ω^* such that

$$J_{\omega^*}(y_0) = \inf_{\omega} J_{\omega}(y_0)$$

(2). The conclusion is still valid if for all initial state y_0 and control policy ω ,

$$\Pr\{Y_t = y^* \text{ for some } t \geq 1 | Y_0 = y_0\} = 1$$

Now the problem is how to compute the optimal control policy. Using the so-called method of successive approximation as follows can do this. Let $\{v_0(y), y \in S - \{y^*\}\}$ be arbitrary, and define

$$v_{n+1}(y) = \min_a \left\{ W_y(a) + \sum_{x \in S - \{y^*\}} q_{yx}(a) v_n(x) \right\}, \quad (3.2) \\ y \in S - \{y^*\}$$

where $q_{yx}(a)$ denotes the probability that the system is moved from state y to state x by action a . Then we have the following lemma [5].

Lemma 3.2 The optimal control policy ω^* for the first-passage problem is determined by

$$J_{\omega^*}(y_0) = \lim_{n \rightarrow \infty} v_{n+1}(y_0) \\ \text{for arbitrary } \{v_0(y), y \in S - \{y^*\}\}$$

3.2. The CMC Approach

Up to this point we can see that Figure 2.2 actually defines a special controlled Markov chain and the theory of controlled Markov chains can be used to determine the required optimal testing strategy. Following Lemma 3.1, we immediately conclude that there exists a deterministic stationary control policy (testing strategy) that minimizes $J_\omega(N)$ (equation (2.1)). To compute the deterministic stationary control policy (testing strategy), we follow the method of successive approximation as indicated in Lemma 3.2.

Suppose the software under test is in state $Y_t = j$, $j \in \{N, N-1, \dots, 2\}$. The optimal action A_t that should be taken at time t can be determined by the following recursive formula

$$v_{n+1}(j) = \min_{1 \leq i \leq m} \left\{ w_{Y_t}(i) + \sum_{k \neq 0} q_{jk}(i) v_n(k) \right\} \\ ; n = 0, 1, 2, \dots$$

where $q_{jk}(i)$ denotes the probability that action i transitions the software from state j to state k , and $\{v_0(k); k = 1, 2, \dots, N\}$ are arbitrary.

Let

$$\lim_{n \rightarrow \infty} v_{n+1}(j) = v(j)$$

We have the following conclusion [2]

$$v(1) = \min_{1 \leq i \leq m} \left\{ \frac{w_1(i)}{\theta_i} \right\} \quad (3.3)$$

$$v(j) = \min_{1 \leq i \leq m} \left\{ \frac{w_j(i)}{j\theta_i} + v(j-1) \right\}; j = 2, 3, \dots, N \quad (3.4)$$

The above equations imply that in state $Y_t \neq 0$, the optimal action that should be taken is the one that

minimizes $\frac{w_{Y_t}(i)}{\theta_i}$ with respect to i .

Example 3.1 Let $N = 3$. Suppose the action set consists of four actions, $A = \{1, 2, 3, 4\}$, and the corresponding defect detection probabilities are

$$\theta_1 = 0.5, \quad \theta_2 = 0.3, \quad \theta_3 = 0.2, \quad \theta_4 = 0.1$$

Further,

$$\begin{aligned}
w_3(1) &= 9, & w_3(2) &= 7, & w_3(3) &= 4, & w_3(4) &= 3 \\
w_2(1) &= 25, & w_2(2) &= 10, & w_2(3) &= 8, & w_2(4) &= 5 \\
w_1(1) &= 35, & w_1(2) &= 17, & w_1(3) &= 10, & w_1(4) &= 8 \\
w_0(1) &= 0, & w_0(2) &= 0, & w_0(3) &= 0, & w_0(4) &= 0
\end{aligned}$$

Then by using equations (3.3) and (3.4), we conclude that in state $Y_i = 3$, action 3 should be taken. In state $Y_i = 2$, action 2 should be taken. In state $Y_i = 1$, action 1 should be taken.

A special case is that $w_{Y_i}(i)$ does not depend on Y_i ,

or $w_{Y_i}(i) \equiv w(i)$ for $Y_i \neq 0$, then in all the non-target states, the optimal action is always the one that minimizes $w(i)/\theta(i)$ with respect to i . We should devote all the testing resources to a single action.

In general in different states different actions might be taken since $w_{Y_i}(i)$ depends on Y_i . A_t thus switches from one action to another, and $\{A_t, t = 0, 1, 2, \dots\}$ defines a test profile of software. For the optimal control policy (testing strategy), since it is deterministic stationary, the resulting $\{Y_t, t = 0, 1, 2, \dots\}$ definitely makes up a Markov chain. However things are different for $\{A_t, t = 0, 1, 2, \dots\}$ which is not necessarily Markovian.

4. Optimal Test Profile

Under optimal testing strategy, Figure 2.2 is converted into Figure 4.1. The software begins with state N (i.e., $Y_0 = N$), and the state transition behavior defines a homogeneous Markov chain. The question of interest here is whether the corresponding $\{A_t, t = 0, 1, 2, \dots\}$ is a Markov chain too. We ignore the trivial case that only one identical action is taken for all the software states, or in another word, A_t has only one state. Another trivial case is that A_t has one-to-one correspondence with Y_t , that is, no identical action is taken for two different software states. Then $\{A_t, t = 0, 1, 2, \dots\}$ is certainly Markovian. In general cases, some different software states correspond to an identical action, some different software states require different actions. A_t is a deterministic function of Y_t . In order to examine behavior of $\{A_t, t = 0, 1, 2, \dots\}$, we distinguish several cases in the rest of this section. Note that Figure 4.1 represents a reducible and homogenous Markov chain. This is different from the often encountered cases that $\{Y_t, t = 0, 1, 2, \dots\}$ is an irreducible and homogenous and

A_t is a deterministic or uncertain function of Y_t , for which the behavior of $\{A_t, t = 0, 1, 2, \dots\}$ has been studied in the mathematical literature [7, 8].

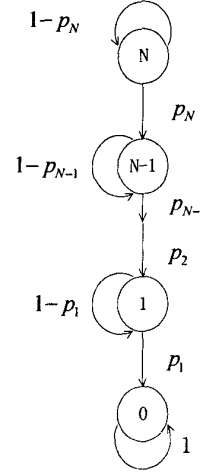


Figure 4.1. Software State Transitions under Optimal Testing Strategy ($p_j; j = 1, 2, \dots, N$).

4.1. Case 1

In this case software states N and $N-1$ correspond to an identical action, and all the other software states have each one-to-one correspondence with actions. Simply, we can say that the original states N and $N-1$ are aggregated into a hyperstate $B = \{N, N-1\}$ and the other states remain unchanged. In the mathematical language, our question is as follows.

Question 4.1 Suppose $\{X_t, t = 0, 1, 2, \dots; X_0 = N\}$ is represented by Figure 4.1 with state space $\{N, N-1, N-2, \dots, 1, 0\}$, $\{X_t, t = 0, 1, 2, \dots; X_0 = N\}$ is still Markovian when the state space is transformed into $\{B, N-2, \dots, 1, 0\}$ with $B = \{N, N-1\}$?

Proposition 4.1 The answer to Question 4.1 is affirmative for $N = 2$.

Proof We can consider different cases one by one to reach the required conclusion.

$$(1). \quad \Pr\{X_{n+1} = 0 | X_n \in B, X_{n-1} \in B, \dots, X_0 \in B\}$$

$$\begin{aligned}
& \Pr\{X_{n-1} \in B, \dots, X_0 \in B | X_{n+1} = 0, X_n \in B\} \\
&= \frac{\Pr\{X_{n+1} = 0, X_n \in B\}}{\Pr\{X_{n-1} \in B, \dots, X_0 \in B | X_n \in B\} \Pr\{X_n \in B\}} \\
&= \Pr\{X_{n+1} = 0 | X_n \in B\}
\end{aligned}$$

This is because

$$\begin{aligned}
& \Pr\{X_{n-1} \in B, \dots, X_0 \in B | X_{n+1} = 0, X_n \in B\} = 1 \\
& \Pr\{X_{n-1} \in B, \dots, X_0 \in B | X_n \in B\} = 1
\end{aligned}$$

$$\begin{aligned}
(2). \quad & \Pr\{X_{n+1} \in B | X_n \in B, X_{n-1} \in B, \dots, X_0 \in B\} \\
&= 1 - \Pr\{X_{n+1} = 0 | X_n \in B, X_{n-1} \in B, \dots, X_0 \in B\} \\
&= \Pr\{X_{n+1} \in B | X_n \in B\}
\end{aligned}$$

$$\begin{aligned}
(3). \quad & \Pr\left\{X_{n+1} = 0 \left| \begin{array}{l} X_n = 0, \dots, X_{n-k} = 0, \\ X_{n-k-1} \in B, \dots, X_0 \in B \end{array} \right. \right\} \\
&= \frac{\Pr\left\{X_{n-k-2} \in B, \dots, X_0 \in B \left| \begin{array}{l} X_{n+1} = 0, X_n = 0, \\ \dots, X_{n-k-1} \in B \end{array} \right. \right\}}{\Pr\{X_{n+1} = 0, X_n = 0, \dots, X_{n-k-1} \in B\}} \\
&= \frac{\Pr\left\{X_{n-k-2} \in B, \dots, X_0 \in B \left| \begin{array}{l} X_n = 0, \dots, \\ X_{n-k-1} \in B \end{array} \right. \right\}}{\Pr\{X_n = 0, \dots, X_{n-k-1} \in B\}} \\
&\quad \times \Pr\{X_n = 0, \dots, X_{n-k-1} \in B\} \\
&= \frac{\Pr\{X_{n+1} = 0, X_n = 0, \dots, X_{n-k-1} \in B\}}{\Pr\{X_n = 0, \dots, X_{n-k-1} \in B\}} \\
&= \frac{\Pr\{X_{n+1} = 0, X_n = 0, \dots, X_{n-k-1} = 1\}}{\Pr\{X_n = 0, \dots, X_{n-k-1} = 1\}} \\
&= \Pr\{X_{n+1} = 0 | X_n = 0, \dots, X_{n-k} = 0, X_{n-k-1} \in B\} \\
&= \Pr\{X_{n+1} = 0 | X_n = 0\}
\end{aligned}$$

$$(4). \quad \Pr\left\{X_{n+1} \in B \left| \begin{array}{l} X_n = 0, \dots, X_{n-k} = 0, \\ X_{n-k-1} \in B, X_0 \in B \end{array} \right. \right\}$$

$$= 0 = \Pr\{X_{n+1} \in B | X_n = 0\}$$

We conclude that $\{X_t, t = 0, 1, 2, \dots; X_0 = N\}$ is Markovian with respect to the state space $\{B, 0\}$.

Q.E.D.

However we should note that $\{X_t, t = 0, 1, 2, \dots; X_0 = N\}$ is no longer homogenous.

In fact,

$$\Pr\{X_{n+1} = 0 | X_n \in B\} = \frac{p_1 \Pr\{X_n = 1\}}{\Pr\{X_n = 1\} + \Pr\{X_n = 2\}}$$

whereas

$$\Pr\{X_n = 1\} = \frac{p_2 [(1 - p_1)^n - (1 - p_2)^n]}{p_2 - p_1}$$

$$\Pr\{X_n = 2\} = (1 - p_2)^n$$

Proposition 4.2 The answer to Question 4.1 is affirmative for $N = 3$.

$$\text{Proof (1). } \Pr\{X_{n+1} \in B | X_n \in B, \dots, X_0 \in B\}$$

$$= \Pr\{X_{n+1} \in B | X_n \in B\}$$

$$(2). \quad \Pr\{X_{n+1} = 1 | X_n \in B, \dots, X_0 \in B\}$$

$$\begin{aligned}
& \Pr\{X_{n-1} \in B, \dots, X_0 \in B | X_{n+1} = 1, X_n \in B\} \\
&= \frac{\Pr\{X_{n+1} = 1, X_n \in B\}}{\Pr\{X_{n-1} \in B, \dots, X_0 \in B | X_n \in B\} \Pr\{X_n \in B\}} \\
&= \frac{\Pr\{X_{n+1} = 1, X_n \in B\}}{\Pr\{X_n \in B\}} = \Pr\{X_{n+1} = 1 | X_n \in B\}
\end{aligned}$$

$$(3). \quad \Pr\left\{X_{n+1} = 1 \left| \begin{array}{l} X_n = 1, \dots, X_{n-k} = 1, \\ X_{n-k-1} \in B, \dots, X_0 \in B \end{array} \right. \right\}$$

$$\begin{aligned}
&= \frac{\Pr\{X_{n+1} = 1, X_n = 1, \dots, X_{n-k} = 1, X_{n-k-1} \in B\}}{\Pr\{X_n = 1, \dots, X_{n-k} = 1, X_{n-k-1} \in B\}} \\
&= \Pr\{X_{n+1} = 1 | X_n = 1, \dots, X_{n-k} = 1, X_{n-k-1} = 2\} \\
&= \Pr\{X_{n+1} = 1 | X_n = 1\}
\end{aligned}$$

$$(4). \quad \Pr\{X_{n+1} = 0 | X_n \in B, \dots, X_0 \in B\} = 0$$

$$= \Pr\{X_{n+1} = 0 | X_n \in B\}$$

$$(5). \quad \Pr\left\{X_{n+1} = 0 \left| \begin{array}{l} X_n = 1, \dots, X_{n-k} = 1, \\ X_{n-k-1} \in B, \dots, X_0 \in B \end{array} \right. \right\}$$

$$= \frac{\Pr\{X_{n+1} = 0, X_n = 1, \dots, X_{n-k} = 1, X_{n-k-1} = 2\}}{\Pr\{X_n = 1, \dots, X_{n-k} = 1, X_{n-k-1} = 2\}}$$

$$= \Pr\{X_{n+1} = 0 | X_n = 1\}$$

$$\begin{aligned}
& \Pr \left\{ X_{n+1} = 0 \left| \begin{array}{l} X_n = 0, \dots, X_{n-k} = 0, X_{n-k-1} = 1, \\ \dots, X_{n-j} = 1, X_{n-j-1} \in B, \dots, X_0 \in B \end{array} \right. \right\} \\
& \Pr \left\{ \begin{array}{l} X_{n+1} = 0, X_n = 0, \dots, X_{n-k} = 0, X_{n-k-1} = 1, \\ \dots, X_{n-j} = 1, X_{n-j-1} \in B \end{array} \right\} \\
(6). &= \frac{\Pr \left\{ \begin{array}{l} X_n = 0, \dots, X_{n-k} = 0, X_{n-k-1} = 1, \\ \dots, X_{n-j} = 1, X_{n-j-1} \in B \end{array} \right\}}{\Pr \left\{ \begin{array}{l} X_n = 0, \dots, X_{n-k} = 0, X_{n-k-1} = 1, \\ \dots, X_{n-j} = 1, X_{n-j-1} \in B \end{array} \right\}} \\
&= \Pr \{X_{n+1} = 0 | X_n = 0\}
\end{aligned}$$

Q.E.D.

Proposition 4.3 The answer to Question 4.1 is affirmative for $N \geq 2$.

Proof We only consider one general case here. Other simple cases can be handled similarly as those in the proofs of Propositions 4.1 and 4.2. For $j = N-2, N-3, \dots, 1, 0$, we have

$$\begin{aligned}
& \Pr \left\{ X_{n+1} = j \left| \begin{array}{l} X_n = j, \dots, X_{n_{j-1}+1} = j, X_{n_{j-1}} = j+1, \\ \dots, X_{n_{j-2}+1} = j+1, X_{n_{j-2}} = j+2, \dots, \\ X_{n_{N-2}+1} = N-3, \dots, X_{n_{N-2}} = N-2, \\ X_{n_{N-2}-1} \in B, \dots, X_0 \in B \end{array} \right. \right\} \\
& \Pr \left\{ \begin{array}{l} X_{n+1} = j, X_n = j, \dots, X_{n_{j-1}+1} = j, X_{n_{j-1}} = j \\ +1, \dots, X_{n_{j-2}+1} = j+1, X_{n_{j-2}} = j+2, \dots, \\ X_{n_{N-2}+1} = N-3, \dots, X_{n_{N-2}} = N-2, \\ X_{n_{N-2}-1} = N-1 \end{array} \right\} \\
&= \frac{\Pr \left\{ \begin{array}{l} X_n = j, \dots, X_{n_{j-1}+1} = j, X_{n_{j-1}} = j+1, \dots, \\ X_{n_{j-2}+1} = j+1, X_{n_{j-2}} = j+2, \dots, X_{n_{N-2}+1} = \\ N-3, \dots, X_{n_{N-2}} = N-2, X_{n_{N-2}-1} = N-1 \end{array} \right\}}{\Pr \left\{ \begin{array}{l} X_n = j, \dots, X_{n_{j-1}+1} = j, X_{n_{j-1}} = j+1, \dots, \\ X_{n_{j-2}+1} = j+1, X_{n_{j-2}} = j+2, \dots, X_{n_{N-2}+1} = \\ N-3, \dots, X_{n_{N-2}} = N-2, X_{n_{N-2}-1} = N-1 \end{array} \right\}} \\
&= \Pr \{X_{n+1} = j | X_n = j\}
\end{aligned}$$

Q.E.D.

Proposition 4.4 Proposition 4.3 is still valid if the state space is replaced by $\{B, j-1, j-2, \dots, 1, 0\}$ with $B = \{N, N-1, \dots, j\}$.

Proof Trivial by using Proposition 4.3 recursively. Q.E.D.

4.2. Case 2

We consider the following question that can be viewed as dual to Question 4.1.

Question 4.2 Suppose $\{X_t, t=0, 1, 2, \dots; X_0 = N\}$ is represented by Figure 4.1 with state space $\{N, N-1, N-2, \dots, 1, 0\}$, is $\{X_t, t=0, 1, 2, \dots; X_0 = N\}$

still Markovian when the state space is transformed into $\{N-1, N-2, \dots, 3, 2, B_0\}$ with $B_0 = \{1, 0\}$?

Proposition 4.5 The answer to Question 4.2 is affirmative for $N = 2$.

Proof (1). $\Pr\{X_{n+1} = 2 | X_n = 2, \dots, X_0 = 2\}$

$$= \Pr\{X_{n+1} = 2 | X_n = 2\}$$

$$(2). \Pr\{X_{n+1} \in B_0 | X_n = 2, \dots, X_0 = 2\}$$

$$\begin{aligned}
&= \Pr \{X_{n+1} = 1 | X_n = 2, \dots, X_0 = 2\} \\
&+ \Pr \{X_{n+1} = 0 | X_n = 2, \dots, X_0 = 2\} \\
&= \Pr \{X_{n+1} \in B_0 | X_n = 2\}
\end{aligned}$$

$$(3). \Pr \left\{ X_{n+1} \in B_0 \left| \begin{array}{l} X_n \in B_0, \dots, X_{n-k} \in B_0, \\ X_{n-k-1} = 2, \dots, X_0 = 2 \end{array} \right. \right\}$$

$$\begin{aligned}
&= \frac{\Pr \left\{ \begin{array}{l} X_{n+1} \in B_0, X_n \in B_0, \dots, X_{n-k} \in B_0, \\ X_{n-k-1} = 2, \dots, X_0 = 2 \end{array} \right\}}{\Pr \{X_n \in B_0, \dots, X_{n-k} \in B_0, X_{n-k-1} = 2, \dots, X_0 = 2\}} \\
&= \frac{\Pr \left\{ \begin{array}{l} X_{n+1} \in B_0, X_n \in B_0, \\ \dots, X_{n-k} \in B_0, X_{n-k-1} = 2 \end{array} \right\}}{\Pr \left\{ \begin{array}{l} X_{n+1} \in B_0, X_n \in B_0, \\ \dots, X_{n-k} \in B_0, X_{n-k-1} = 2 \end{array} \right\}} \\
&\times \Pr \left\{ \begin{array}{l} X_{n+1} \in B_0, X_n \in B_0, \\ \dots, X_{n-k} \in B_0, X_{n-k-1} = 2 \end{array} \right\} \\
&= \frac{\Pr \left\{ \begin{array}{l} X_{n+1} \in B_0, X_n \in B_0, \dots, X_{n-k+1} \in B_0, \\ X_{n-k} = 1, X_{n-k-1} = 2 \end{array} \right\}}{\Pr \{X_n \in B_0, \dots, X_{n-k} \in B_0, X_{n-k} = 1, X_{n-k-1} = 2\}} \\
&= \frac{\Pr \left\{ \begin{array}{l} X_{n+1} \in B_0, X_n \in B_0, \dots, X_{n-k+1} \in B_0, \\ X_{n-k} = 1, X_{n-k-1} = 2 \end{array} \right\}}{\Pr \{X_n \in B_0, \dots, X_{n-k+1} \in B_0, X_{n-k} = 1\}} \\
&= \frac{\Pr \{X_{n+1} \in B_0, X_n \in B_0, \dots, X_{n-k+1} \in B_0, X_{n-k} = 1\}}{\Pr \{X_n \in B_0, \dots, X_{n-k+1} \in B_0, X_{n-k} = 1\}} \\
&= 1 = \Pr \{X_{n+1} \in B_0 | X_n \in B_0\}
\end{aligned}$$

This is because

$$\begin{aligned}
& \Pr\{X_{n+1} \in B_0, X_n \in B_0, \dots, X_{n-k+1} \in B_0, X_{n-k} = 1\} \\
&= \Pr\{X_{n+1} = 1, X_n \in B_0, \dots, X_{n-k+1} \in B_0, X_{n-k} = 1\} \\
&+ \Pr\{X_{n+1} = 0, X_n \in B_0, \dots, X_{n-k+1} \in B_0, X_{n-k} = 1\} \\
&= \Pr\{X_{n+1} = 1, X_n = 1, \dots, X_{n-k+1} = 1, X_{n-k} = 1\} \\
&+ \Pr\left\{X_{n+1} = 0, X_n = 1, X_{n-1} \in B_0, \dots, \right. \\
&\quad \left. X_{n-k+1} \in B_0, X_{n-k} = 1\right\} \\
&+ \Pr\left\{X_{n+1} = 0, X_n = 0, X_{n-1} \in B_0, \dots, \right. \\
&\quad \left. X_{n-k+1} \in B_0, X_{n-k} = 1\right\} \\
&= \Pr\{X_{n+1} = 1, X_n = 1, \dots, X_{n-k+1} = 1, X_{n-k} = 1\} \\
&+ \Pr\{X_{n+1} = 0, X_n = 1, \dots, X_{n-k+1} = 1, X_{n-k} = 1\} \\
&+ \sum_{x_{n-1}, \dots, x_{n-k+1} \in B_0} \Pr\left\{X_{n+1} = 0, X_n = 0, X_{n-1} = x_{n-1}, \dots, \right. \\
&\quad \left. \dots, X_{n-k+1} = x_{n-k+1}, X_{n-k} = 1\right\} \\
&= \Pr\{X_n = 1, \dots, X_{n-k+1} = 1, X_{n-k} = 1\} + \\
&\quad \sum_{x_{n-1}, \dots, x_{n-k+1} \in B_0} \Pr\left\{X_{n+1} = 0 \left| \begin{array}{l} X_n = 0, X_{n-1} = x_{n-1}, \dots, \\ X_{n-k+1} = x_{n-k+1}, X_{n-k} = 1 \end{array} \right. \right\} \\
&\quad \times \Pr\left\{X_n = 0, X_{n-1} = x_{n-1}, \dots, \right. \\
&\quad \left. X_{n-k+1} = x_{n-k+1}, X_{n-k} = 1\right\} \\
&= \Pr\{X_n = 1, X_{n-1} \in B_0, \dots, X_{n-k+1} \in B_0, X_{n-k} = 1\} \\
&+ \sum_{x_{n-1}, \dots, x_{n-k+1} \in B_0} \Pr\left\{X_n = 0, X_{n-1} = x_{n-1}, \dots, \right. \\
&\quad \left. X_{n-k+1} = x_{n-k+1}, X_{n-k} = 1\right\} \\
&= \Pr\{X_n = 1, X_{n-1} \in B_0, \dots, X_{n-k+1} \in B_0, X_{n-k} = 1\} \\
&+ \Pr\{X_n = 0, X_{n-1} \in B_0, \dots, X_{n-k+1} \in B_0, X_{n-k} = 1\} \\
&= \Pr\{X_n \in B_0, X_{n-1} \in B_0, \dots, X_{n-k+1} \in B_0, X_{n-k} = 1\}
\end{aligned}$$

Q.E.D.

Proposition 4.6 The answer to Question 4.2 is affirmative for $N = 3$.

Proof We only consider one case. The other cases are easy to verify. There holds

$$\begin{aligned}
& \Pr\left\{X_{n+1} \in B_0 \left| \begin{array}{l} X_n \in B_0, \dots, X_{n-k} \in B_0, X_{n-k-1} = 2, \\ \dots, X_{n-j} = 2, X_{n-j-1} = 3, \dots, X_0 = 3 \end{array} \right. \right\} \\
&= \frac{\Pr\left\{X_{n+1} \in B_0, X_n \in B_0, \dots, X_{n-k} \in B_0, \right. \\
&\quad \left. X_{n-k-1} = 2, \dots, X_{n-j} = 2, X_{n-j-1} = 3\right\}}{\Pr\left\{X_n \in B_0, \dots, X_{n-k} \in B_0, \right. \\
&\quad \left. X_{n-k-1} = 2, \dots, X_{n-j} = 2, X_{n-j-1} = 3\right\}} \\
&= 1 = \Pr\{X_{n+1} \in B_0 | X_n \in B_0\}
\end{aligned}$$

This is because

$$\begin{aligned}
& \Pr\left\{X_{n+1} \in B_0, X_n \in B_0, \dots, X_{n-k} \in B_0, \right. \\
&\quad \left. X_{n-k-1} = 2, \dots, X_{n-j} = 2, X_{n-j-1} = 3\right\} \\
&= \Pr\left\{X_{n+1} = 1, X_n \in B_0, \dots, X_{n-k+1} \in B_0, \right. \\
&\quad \left. X_{n-k} = 1, X_{n-k-1} = 2, \dots, X_{n-j} = 2, X_{n-j-1} = 3\right\} \\
&+ \Pr\left\{X_{n+1} = 0, X_n \in B_0, \dots, X_{n-k+1} \in B_0, \right. \\
&\quad \left. X_{n-k} = 1, X_{n-k-1} = 2, \dots, X_{n-j} = 2, X_{n-j-1} = 3\right\} \\
&= \Pr\left\{X_{n+1} = 1, X_n = 1, \dots, X_{n-k+1} = 1, X_{n-k} = 1, \right. \\
&\quad \left. X_{n-k-1} = 2, \dots, X_{n-j} = 2, X_{n-j-1} = 3\right\} \\
&+ \Pr\left\{X_{n+1} = 0, X_n = 1, \dots, X_{n-k+1} = 1, X_{n-k} = 1, \right. \\
&\quad \left. X_{n-k-1} = 2, \dots, X_{n-j} = 2, X_{n-j-1} = 3\right\} \\
&+ \Pr\left\{X_{n+1} = 0, X_n = 0, X_{n-1} \in B_0, \dots, X_{n-k+1} \in B_0, \right. \\
&\quad \left. X_{n-k} = 1, X_{n-k-1} = 2, \dots, X_{n-j} = 2, X_{n-j-1} = 3\right\} \\
&= \Pr\left\{X_n = 1, \dots, X_{n-k+1} = 1, X_{n-k} = 1, X_{n-k-1} = 2, \right. \\
&\quad \left. \dots, X_{n-j} = 2, X_{n-j-1} = 3\right\} \\
&+ \Pr\left\{X_n = 0, X_{n-1} \in B_0, \dots, X_{n-k+1} \in B_0, \right. \\
&\quad \left. X_{n-k} = 1, X_{n-k-1} = 2, \dots, X_{n-j} = 2, X_{n-j-1} = 3\right\} \\
&= \Pr\left\{X_n \in B_0, X_{n-1} \in B_0, \dots, X_{n-k+1} \in B_0, \right. \\
&\quad \left. X_{n-k} = 1, X_{n-k-1} = 2, \dots, X_{n-j} = 2, X_{n-j-1} = 3\right\}
\end{aligned}$$

Q.E.D.

Proposition 4.7 The answer to Question 4.2 is affirmative for $N \geq 2$.

Proof For $N > 3$ the proof is similar to that of Proposition 4.6. Q.E.D.

Proposition 4.8 Proposition 4.7 is still valid if the state space is replaced by $\{N, N-1, \dots, j, B_0\}$ with $B_0 = \{j-1, j-2, \dots, 1, 0\}$.

Proof Trivial by using Proposition 4.7 recursively.

Q.E.D.

4.3. Case 3

Consider Figure 4.1 again. Let $N=3$ and $B=\{1,2\}$. That is, the state space is transformed into $\{3, B, 0\}$. Note

$$\begin{aligned}
& \Pr\{X_{n+1} = 0 | X_n \in B, X_{n-1} = 3, \dots, X_0 = 3\} \\
&= \frac{\Pr\{X_{n+1} = 0, X_n = 2, X_{n-1} = 3, \dots, X_0 = 3\} \\
&\quad + \Pr\{X_{n+1} = 0, X_n = 1, X_{n-1} = 3, \dots, X_0 = 3\}}{\Pr\{X_n = 2, X_{n-1} = 3, \dots, X_0 = 3\} \\
&\quad + \Pr\{X_n = 1, X_{n-1} = 3, \dots, X_0 = 3\}} \\
&= 0
\end{aligned}$$

However

$$\Pr\{X_{n+1}=0|X_n \in B\} = \frac{p_1 \times \Pr\{X_n=1\}}{\Pr\{X_n=1\} + \Pr\{X_n=2\}} \neq 0 \quad \text{So}$$

$\{X_t, t=0,1,2,\dots; X_0=3\}$ is no longer Markovian with respect to the state space $\{3, B, 0\}$.

4.4. Case 4

For $N=3$, suppose $B=\{1,3\}$. We have

$$\begin{aligned} & \Pr\{X_{n+1}=2|X_n \in B, X_{n-1} \in B\} \\ &= \frac{(1-p_3)p_3 \Pr\{X_{n-1}=3\}}{(1-p_3)\Pr\{X_{n-1}=3\} + (1-p_1)\Pr\{X_{n-1}=1\}} \neq 0 \end{aligned}$$

However

$$\begin{aligned} & \Pr\{X_{n+1}=2|X_n \in B, X_{n-1}=2\} \\ & \Pr\{X_{n+1}=2, X_n=1, X_{n-1}=2\} + \\ &= \frac{\Pr\{X_{n+1}=2, X_n=3, X_{n-1}=2\}}{\Pr\{X_n=1, X_{n-1}=3\}} = 0 \end{aligned}$$

So $\{X_t, t=0,1,2,\dots; X_0=3\}$ is no longer Markovian with respect to the state space $\{B, 2, 0\}$.

4.5. Case 5

Again for $N=3$, let $B=\{0,2\}$. We have

$$\begin{aligned} & \Pr\{X_{n+1} \in B|X_n \in B, X_{n-1}=3\} \\ &= \frac{\Pr\{X_{n+1}=2, X_n=2, X_{n-1}=3\}}{\Pr\{X_n=2, X_{n-1}=3\}} = 1 - p_2 \end{aligned}$$

However

$$\begin{aligned} & \Pr\{X_{n+1} \in B|X_n \in B, X_{n-1} \in B\} \\ &= \frac{\Pr\{X_{n-1}=0\} + (1-p_2)^2 \Pr\{X_{n-1}=2\}}{\Pr\{X_{n-1}=0\} + (1-p_2)\Pr\{X_{n-1}=2\}} \neq 1 - p_2 \end{aligned}$$

Suppose $p_2 \neq 0$. So $\{X_t, t=0,1,2,\dots; X_0=3\}$ is no longer Markovian with respect to the state space $\{3, 1, B\}$.

4.6. Case 6

Let $N=4$ and $B=\{1,3\}$. That is, we consider the state space $\{4, B, 2, 0\}$. Note

$$\begin{aligned} & \Pr\{X_{n+1}=2|X_n \in B, X_{n-1}=4\} \\ &= \frac{\Pr\{X_{n+1}=2, X_n=3, X_{n-1}=4\}}{\Pr\{X_n=3, X_{n-1}=4\}} = p_3 \end{aligned}$$

However

$$\begin{aligned} & \Pr\{X_{n+1}=2|X_n \in B, X_{n-1} \in B\} \\ &= \frac{p_3(1-p_3)\Pr\{X_{n-1}=3\}}{(1-p_1)\Pr\{X_{n-1}=1\} + (1-p_3)\Pr\{X_{n-1}=3\}} \neq p_3 \end{aligned}$$

So $\{X_t, t=0,1,2,\dots; X_0=4\}$ is no longer Markovian with respect to the state space $\{4, B, 2, 0\}$.

4.7. Conclusion

Up to this point we can conclude that the optimal test profile specifies a nonhomogenous Markov chain if two conditions are satisfied:

- (1). A number of (possibly one) consecutive software states beginning with the initial software state or ending with the absorbing software state correspond to an identical action; and
- (2). For all other software states, each corresponds to an action not taken in other states.

If these conditions are not satisfied, evidence shows that the corresponding optimal test profile is no longer Markovian, although this observation needs further mathematical justification. Here we note that the absorbing software may correspond to an action not taken in any other state. Alternatively, we may say that the action taken in the absorbing state is the same as that taken in its neighbor state. This interpretation will not affect the validity of the above conclusion we draw.

5. Concluding Remarks

In the last sections we have discussed how to treat software testing as a control problem and how the controlled Markov chains (CMC) approach to software testing determines an optimal testing strategy. Under the optimal testing strategy, the software state transitions behave as a Markov chain. However the corresponding optimal test profile is not necessarily Markovian. If a number of consecutive software states beginning with the initial software state or ending with the absorbing software state correspond to an identical action, and for all other states each corresponds to an action not taken in other states, then the corresponding optimal test profile is Markovian. In other cases, evidence shows that the corresponding optimal test profile is not Markovian. The

analyses presented so far deepen our understanding of the CMC approach to software testing. In the meantime, they also provide theoretical justification that under certain circumstance software test profile or operational profile fits a Markov model. We note that homogenous Markov models have been extensively used to characterize software operational profile [9]. The analyses of this paper suggest that nonhomogenous Markov models should also be considered for software operational profile modeling.

References

- [1] K.Y.Cai, "On the Concepts of Total Systems, Total Dependability and Software Cybernetics", (unpublished manuscript), *Centre for Software Reliability, City University, London*, Draft version, October 1994; revised version, 1995.
- [2] K.Y.Cai, "A Controlled Markov Chains Approach to Software Testing", *submitted for publication*, 2000.
- [3] B.Beizer, *Software Testing Techniques (2nd Edition)*, Van Nostrand Reinhold, 1990.
- [4] K.Y.Cai, "Towards a Conceptual Framework of Software Run Reliability Modeling", *Information Sciences*, Vol.126, 2000, pp137-163.
- [5] C.Derman, *Finite State Markovian Decision Processes*, Academic Press, 1970.
- [6] O.Hernandez-Lerma, J.B.Lasserre, *Discrete-Time Markov Control Processes: Basic Optimality Criteria*, Springer, 1996.
- [7] G.Rubino, B.Sericola, "A Finite Characterization of Weak Lumpable Markov Processes. Part I: The Discrete Time Case", *Stochastic Processes and Their Applications*, Vol.38, 1991, pp195-204.
- [8] P.Spreij, "On the Markov Property of a Finite Hidden Markov Chain", *Statistics & Probability Letters*, Vol.52, 2001, pp279-288.
- [9] K.Y.Cai, *Software Defect and Operational Profile Modeling*, Kluwer Academic Publishers, 1998.